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Biotic Stress Management in Soil-Less Agriculture Systems: A Deep Learning Approach for Identification of Leaf Miner Pest Infestation

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Abstract

Leaf miner pests pose a serious threat to the productivity, profitability, and sustainability of soil-less tomato cultivation systems. Early and accurate identification of leaf miner infestation is crucial for timely pest control measures. This study presents an efficient approach using attention-based convolutional neural networks for timely identification of this pest infestation. The proposed approach uses both spatial and channel attention modules to enhance the feature extraction capability of the convolutional neural network. The custom model developed was trained using an image dataset collected from tomatoes grown in a hydroponic environment. The different hyper parameters were tuned to get the optimal model performance. The experimental results show that the proposed attention-based CNN model achieved an overall accuracy of 97.87%, 97.10% precision, 98.53% recall, and 97.81% F1-score. Additionally, the model performance was compared with other pre-trained models viz., AlexNet, VGG16, and VGG19, and was found to outperform these state-of-the-art CNN models due to its improved feature extraction capability. The efficiency of the model underlines its potential to be deployed as part of automated pest monitoring systems in hydroponic environments. This work contributes to the development of computer vision and deep learning-based solutions for precision agriculture applications.

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Keywords: Attention mechanism, CNN, Pest identification, Precision agriculture

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1. Introduction

The growing population and increased urbanization have put unprecedented pressure on the agriculture industry to increase food production [1]. Traditional agriculture relies heavily on arable land, however, due to increased population, the availability of agricultural land is declining at a drastic level. Climate change, land degradation, water shortage, overuse of chemical pesticides, etc. also endanger global food security. Therefore, there is an urgent need to adopt more sustainable and efficient forms of agriculture practices. Soil-less systems, specifically hydroponic systems, have been identified as a sustainable solution for cultivating crops in urban environments, with utmost control on crop growth [2]. Hydroponics is a soil-less crop cultivation technique that grows crops in the presence of a nutrient solution. With precise control over the nutrients, growing conditions, and climate, hydroponic systems can enhance productivity from limited space and amidst land as well as water constraints [3]. Further, these systems provide options to grow crops nearer to urban markets, reducing overall transportation costs.

The biotic stress stemming from living organisms creates a serious danger to plant health and agricultural productivity. Pest infestation is one major challenge to hydroponic cultivation. Pests are mainly classified based on their characteristics and the damage they cause. It mainly includes borers such as tomato fruit borers, chewing insects like beetles, sucking insects such as aphids, mites, etc. In hydroponics cultivation under closed greenhouses, the pests can spread rapidly without any physical barriers. When left unchecked, these pests multiply, leading to entire crop loss. Leaf miners are one major, most common and damaging pest found in horticulture crops such as tomatoes [4]. Leaf miners are the larvae of various fly species, and they burrow into the leaf tissues and cause disruption to the plant's photosynthesis process and overall health. The larvae develop between the surfaces of the leaf and 'mine' out the middle layer, as it matures and indicates a zigzag tunnel like appearance. This causes significant damage to the crop, resulting in huge yield loss. An automatic system that can identify pest infestation can be a feasible solution to overcome the key challenges faced by manual inspection [5]. Mimicking human vision through imaging devices and computational models was one of the most suited approaches for pest identification. The traditional computer vision approaches were focused on hand-crafted image features and shallow machine learning models. The predefined visual features like color, texture, shape, etc were extracted from the images as feature vectors and further they were analyzed using machine learning classification algorithms. The major downside and lack of performance improvement was due to the difficulty in manually extracting the features for identification. Over the past few years, CNNs and their various architectures have been utilized across different domains to automate computer vision tasks with increased precision compared to the traditional hand-crafted feature-based models [6]. The evolution of big data sets, computing power, and cloud storage facilities has made CNN grow deeper and more complex. Integration of approaches like residual connections and batch normalization have given greater efficiency to CNNs. Pre-trained models such as AlexNet, ResNet, Inception, DenseNet, etc. have attained improved results in several computer vision tasks in agriculture. In this study, a custom CNN with channel and spatial attention mechanism was developed for identifying leaf miner infestation in tomato crops.

2. Related Work

Numerous investigations have explored the application of advanced machine learning and deep learning algorithms for addressing the pest infestation identification. The authors of [7] employed ANFIS classifiers for identification of leaf miner infestation and Alternaria disease on tomato leaves. The models were developed on an RGB dataset of 272 tomato leaf images and achieved an accuracy of 87.78% and 85.23% for the segmentation of alternaria and leaf miner pests respectively. The CNN pre-trained models VGG16, VGG19, and ResNet50 also provided state-of-the-art results in identifying leaf miner invasion in tomato crops. Out of these selected models, VGG16 has shown a superior accuracy of 91.9% in pest identification [8]. For the detection of leaf miners in tomato crops, the authors of [9] developed two different CNN models viz., based on ResNet-152 and Mask-RCNN respectively. With respect to the diagnosing performance in terms of precision, recall, and F1-score, the Mask-RCNN based model was found to outperform the classification model. In another study, an ensemble classifier based on CNN was proposed for recognizing pests in citrus leaves [10]. The ensemble approach achieved an overall accuracy of 99.04% when trained on the dataset consisting of 1774 citrus images. Smartphone-based solutions are highly effective for automating the

damage detection caused by the leaf miner pest. A UNet model was proposed in [11] and trained on a dataset consisting of 1212 leaf images. Further, the deployment was deployed as a smartphone-based solution and the model has shown a detection confidence of 70%. Most of the aforementioned approaches in modeling leaf miner pest infestation have utilized pre-trained CNN models and object detection models like Yolo models. These models are trained on large datasets like ImageNet with millions of images and thousands of classes, making the models relatively slower, less flexible and more resource-intensive to run, in comparison to the lightweight custom CNN models. The key contribution of this study is the implementation of a more robust classification model by combining the attention mechanism into a custom CNN model to dynamically focus on the most informative regions of the images instead of the entire input image. In the proposed study, developing a custom attention-based CNN model can offer better control over the learning representations, optimized for the target data distribution. The other contribution of this investigation is the preparation of a new tomato leaf dataset consisting of leaf miner pest-infected and healthy leaf images, captured in a hydroponic environment.

3. Materials and Methods

3.1. Data collection & Pre-processing

In the present study, the data collection was carried out from the hydroponics environment located at Bhopal, Madhya Pradesh, India (longitude - 77° 22′ 36.5″ E and latitude- 23° 17′ 53.3” N). The hydroponic cherry tomatoes were grown using the NFT technique and the images were captured in multiple phases using a smartphone camera (Xiaomi POCO X3, having 64MP with Sony IMX 682). The images were captured between 8 AM to 4 PM inside the hydroponic setup by placing the camera at a distance of approximately 30-40 cm away from the crop. The image collection was carried out in multiple stages starting from 40 days to 80 days after planting. The final dataset contained 2820 RGB images of both healthy and leaf miner-infected tomato leaves, examined and labeled by an agricultural expert. The snapshot of the data collection environment and the dataset is shown in Fig.1. The images collected are pre-processed to optimize the images for further processing. In the initial phase, the captured data were manually inspected to remove the low-quality images. The data normalization and augmentation are applied to the dataset. Different data augmentation techniques like flipping and rotation were applied to introduce the variability in the dataset, enhancing the model's generalization capability to different image variations. For data augmentation, the OpenCV library was used. Cv2.rotate () and cv2.flip () functions in OpenCV were used for the rotation (45, 90 degrees) and flipping of the images. The final dataset was well balanced with 1459 images of healthy leaves and 1361 images of leaf miner infected leaves. The dataset was further divided into train, validation and test sets in 80:10:10 ratio for model training, validation and testing. The complete methodology followed in the study is shown in Fig. 2.

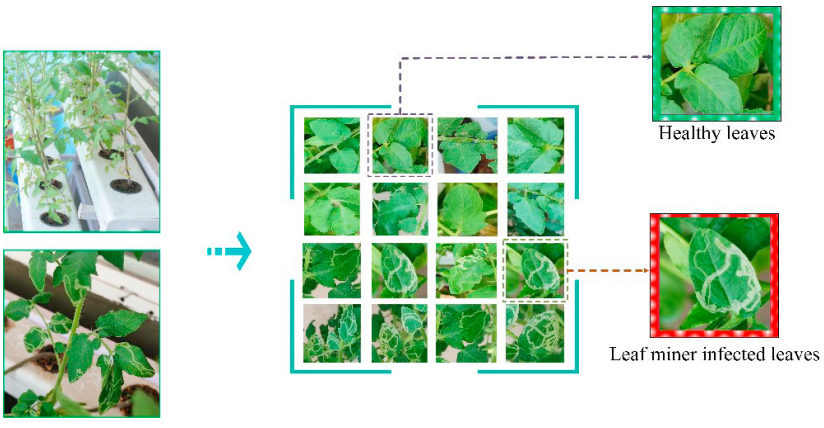


Fig. 1. Snapshot of the data collection environment and image dataset

3.2. Attention Mechanism and CBAM Module

Attention mechanisms in deep learning models can play a crucial role in enhancing prediction accuracy and computational efficiency to a great extent [12]. The attention mechanism helps the model to selectively filter out a small amount of key information, ignoring the unimportant elements in the data [13]. By prioritizing and highlighting the most important features, the overall model performance is improved. This approach is very similar to how the human vision system focuses on a specific object, among the huge set of different objects present in the field of view. CBAM leverages attention mechanism in two dimensions viz., channel and space (Fig. 3). The most significant features are highlighted, and irrelevant features are suppressed. The attention weights are inferred sequentially along with these two dimensions from the intermediate feature map. These values are multiplied with the actual feature map to adaptively adjust and refine the features. Each CBAM block takes the previous convolution layer's feature map as its input and produces a refined feature map as the output, which is then passed to the next layer. Assume FM $\in$ K^{LxHxW} be the input feature map where L indicates the channels, H denotes the height and W denotes the width the feature map holds. The features are passed to the channel attention module to generate the first refined feature map (FM') [14] (Eq. 1). The channel attention module produces a one-dimensional attention map denoted by Mc $\in$ $K^{1\times H\times W}$.

$$FM' = Mc(FM) \oplus FM \quad (1)$$

where $\oplus$ indicates elementwise multiplication. FM' highlights the vital information present in the image. Further, this output is passed into the spatial attention module and generates the final refined feature FM'' (Eq. 2).

$$FM'' = Mc(FM') \oplus FM' \quad (2)$$

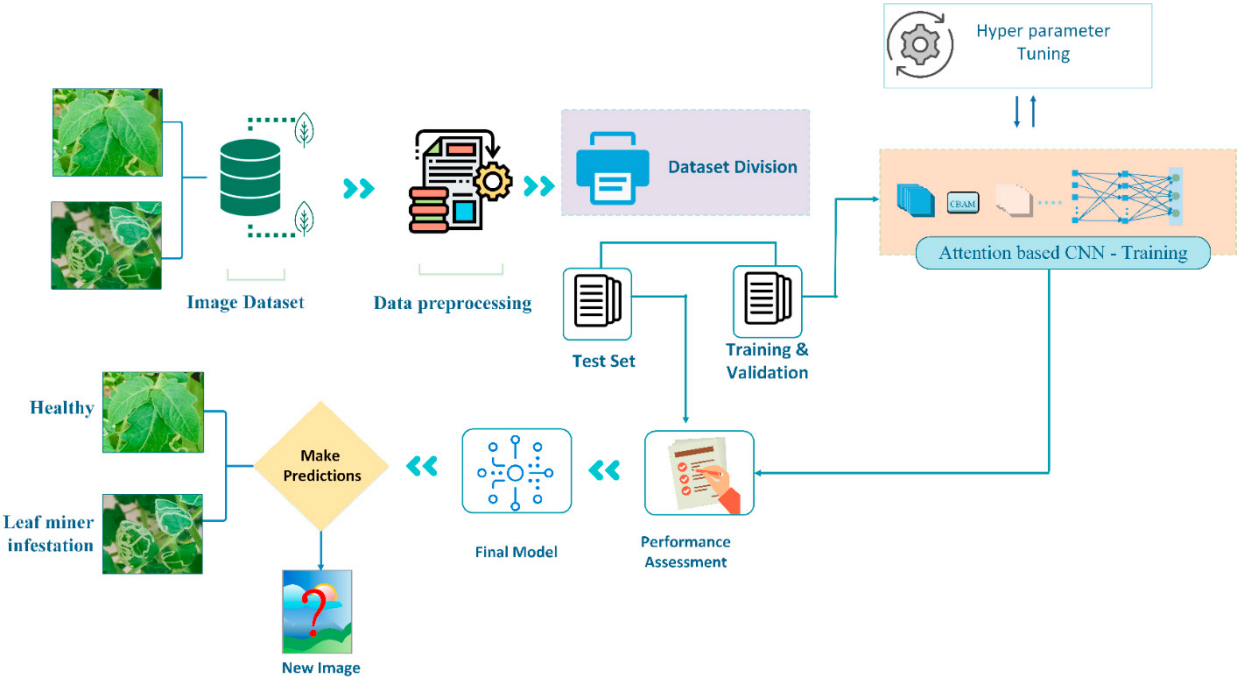


Fig. 2. Snapshot of the data collection environment and image dataset

FM'' refines the feature map further, capturing all relevant features. It is worth noting that the spatial and channel attention structures are mutually independent and this allows them to function as independent modules in the

convolutional neural network architectures. In the present work, a custom CBAM-CNN model is designed as a leaf miner pest infestation identification model, which has improved feature extraction ability and high predictive power.

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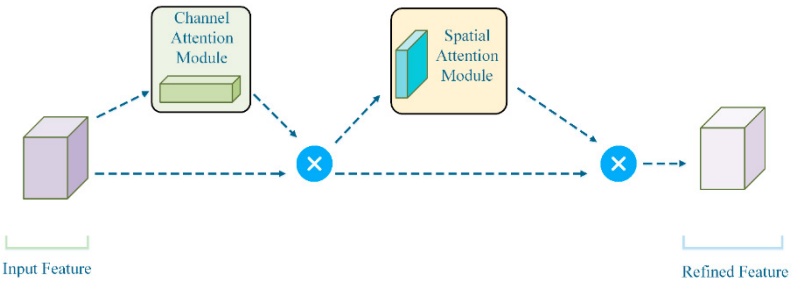


Fig. 3. Convolutional Block Attention Module

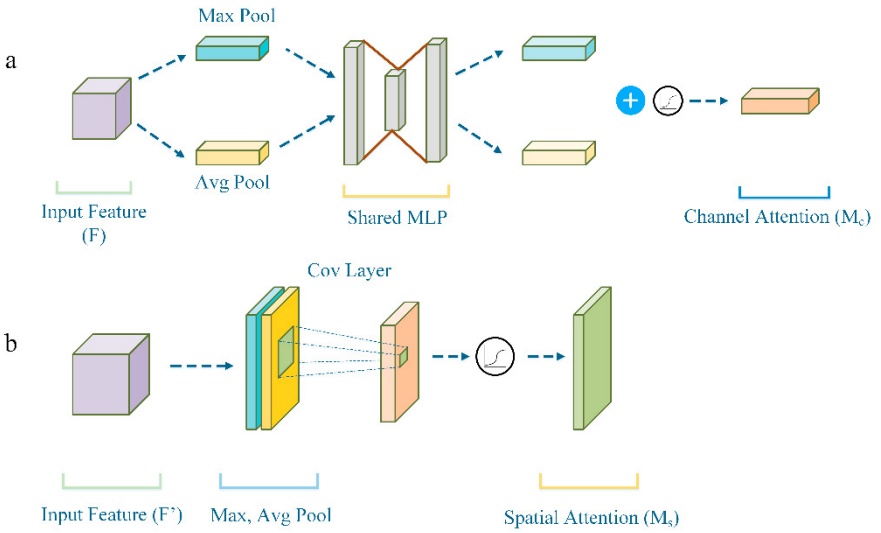


Fig. 4. Components of CBAM (a) Channel attention module (b) Spatial attention module

3.3. The custom CNN Architecture

A custom CNN model was proposed in this study for the identification of leaf miner pest infestation. The model incorporates the channel and spatial attention for improved feature extraction capability. The inputs to the CNN model are the RGB image with 225x225 pixels size. The architecture comprises mainly a series of convolutional layers, maxpooling layers and 3 CBAM blocks to enhance feature extraction. The CBAM structure is shown in Fig. 4. The input images were passed onto the convolutional layer to extract the features. At the beginning, two convolutional layers were used with 32 filters, each with a filter size of 3x3 (Fig. 5). The activation function used in each layer is 'ReLU' and the kernel was initialized using He uniform variance scaling initializer. It draws samples from a uniform distribution based on the number of input units in the weight tensor [15]. The convolutional block attention module (CBM) was added after the convolutional module. The CBAM typically integrates both channel and spatial based attention mechanisms. For effective computation of channel attention, the spatial dimension of the input feature map was squeezed. Average pooling was employed in architecture to aggregate the spatial information. In order to infer finer channel-wise attention, the max-pooling operation is also performed simultaneously. The channel attention map

is generated with the help of a shared network, consisting of multi-layer perceptron with one hidden layer. The output feature vectors are merged using element-wise summation. The spatial map is generated by capturing the inter-spatial relationships. For computing spatial attention, the average pooling followed by max-pooling operation has been applied along the channel axis. Further, these values are concatenated to generate an efficient feature descriptor. A convolutional layer is attached for generating the spatial attention map. In the custom CNN architecture, following the first CBAM, batch normalization and maxpooling layers were added to the model configuration. The batch normalization normalizes the inputs to the layer across mini batches during training and addresses the challenging internal covariance effect that hinders the optimization process and slow convergence. The kernel size for the max-pooling layer was set to 2x2. In the next stage, two convolutional layers with 64 filters each were added. Further, the sequence of CBAM, batch normalization and maxpooling layers were repeated. Additionally, a convolutional layer with 128 filters was also configured. A flattened layer is added to convert the features into linear vectors. Two fully connected layers with 'ReLU' activation have been added to the network. A dropout layer with a rate of 0.2 was used to avoid model overfitting. In the last stage, a dense layer with two output neurons and sigmoid activation was added for the binary classification.

3.4. Model Evaluation

The developed model was evaluated on the test dataset to identify the effectiveness of the model. The present study uses four different evaluation indicators viz., accuracy, precision, recall, and F1- score. The values of each are calculated in percent (%). The formulae for each are shown in Eq. (3-6). TN represents True Negative, FP represents False Positive, TP stands for True Positive and FN shows False Negative.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

$$\mathrm{Pr}ecision = \frac{TP}{TP + FP} \tag{4}$$

$$Sensitivity / \mathrm{Re}call = \frac{TP}{TP + FN} \tag{5}$$

$$F1 - Score = \frac{2 * \mathrm{Pr}ecision * \mathrm{Re}call}{\mathrm{Pr}ecision + \mathrm{Re}call} \tag{6}$$

3.5. Hardware and Software Tools

In the experiment, the models were developed in Python programming language with libraries for image processing and deep learning. The study utilized Google Colab platform that serves high-end CPU and GPU in the cloud environment. NVIDIA Tesla T4 GPU with 16GB VRAM was used to train and evaluate the model. This embellished environment gave provided seamless training for all the models in the study. The different tools and libraries used in the experiment are shown in Table 1.

Table 1. Tools and packages used in the model development environment

Tools/Package	Version
Python	3.10.12
Keras	2.12.0
Pandas	1.5.3
Pillow	9.4.0
Scikit-learn	1.2.2
Matplotlib	3.7.1
Numpy	1.23.5
Opencv-python	4.8.0.76

4. Results and Discussion

To demonstrate the efficiency of the model, several experiments were conducted by varying different parameters related to the attention-based CNN model. In the first stage, the optimal model is found using extensive hyperparameter tuning. For this, the proposed attention-based custom model was trained under different hyperactive parameter configurations of batch size (4,8 and 16), epoch (200), and learning rate (0.01, 0.001). For the seamless training of the model, the dataset was exported into Google Drive and mounted to the Google Colab platform. The developed attention-based DCNN model possesses 103,694,658 total trainable parameters. The model size was observed to be 239 MB. All the models were trained for 200 epochs and two different approaches were used to prevent the overfitting viz., data augmentation and early stopping. During the training process, all the models converged during early epochs without overfitting. The loss function and accuracy graph of the model when trained for 200 epochs, batch size 16 and learning rate 0.001 is given in Fig. 6. The proposed model has shown accuracy 97.87%, precision 97.10%, recall 98.53% and F1-score 97.81%. The overall training time and testing time for the model were 2640 seconds and 540 seconds respectively.

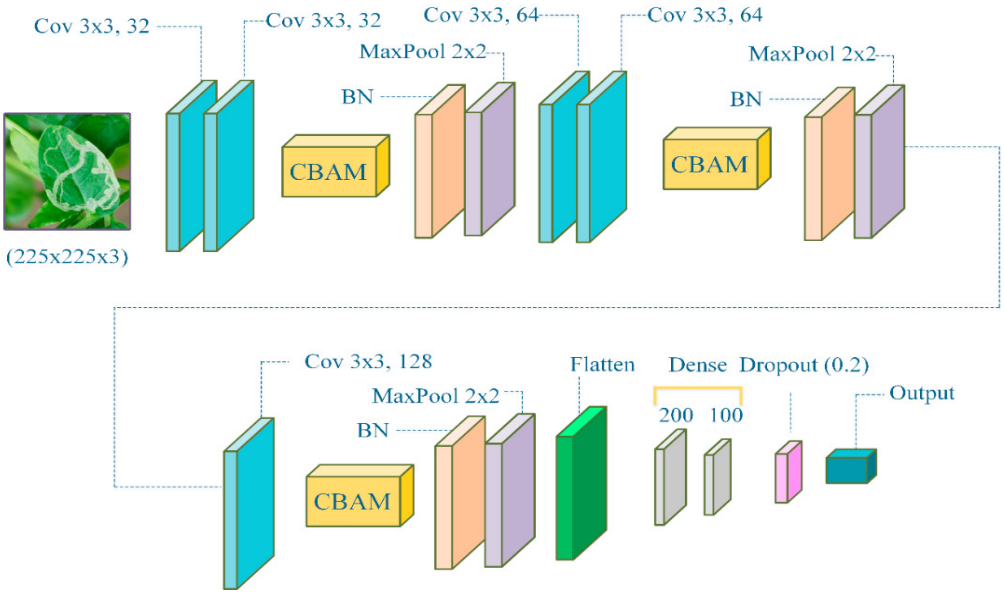


Fig. 5. Convolutional block attention module based Custom CNN architecture

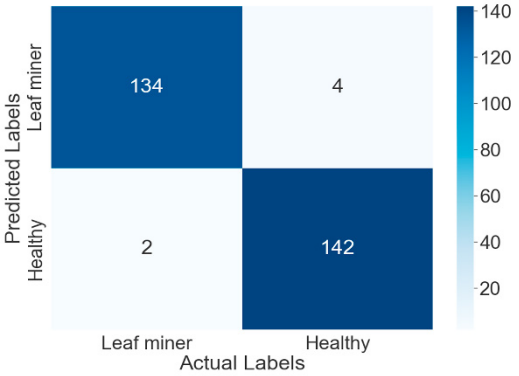


Fig. 6. Confusion matrix of the proposed Model



Fig. 7. Performance plots of the proposed model (a) Loss graph (b) Accuracy graph

To assess the effectiveness of the proposed model compared to other models, we handpicked a few less complex state-of-the-art pre-trained CNN models viz., AlexNet [16], VGG16 [17], and VGG19 [18]. All the models were trained with the same hyperparameter configuration as that of the attention-based custom model. The models were converged without any overfitting; this was ensured through the early stopping criteria set during the training process. The performance metrics were evaluated using a confusion matrix, and the proposed model has clearly outperformed all the other models in terms of accuracy, recall, precision, and F1 score (Fig. 7). Among the chosen pre-trained DCNN models, VGG19 has shown superior performance compared to the other two models with an accuracy of 96.10%, precision of 94.93%, recall of 97.04%, and F1 score of 95.97% (Table 2).

Table 2. Performance comparison of proposed models with AlexNet, VGG16 and VGG19

Model	**Epoch and Stoppage**	**Batch size**	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1 Score (%)**
AlexNet	200/94	16	92.2	90.58	93.28	91.91
VGG16	200/125	8	94.33	92.75	95.52	94.11
VGG19	200/127	16	96.10	94.93	97.04	95.97
Proposed Model	**200/119**	**16**	**97.87**	**97.10**	**98.53**	**97.81**

The lowest performance was observed for AlexNet. For batch size 16, AlexNet has shown accuracy on the test set with a value of 92.2%. The corresponding precision, recall and F1-score values were 90.58%, 93.28% and 91.91% respectively. All the models except VGG16 have shown the highest accuracy when trained with batch size 16. For VGG16, the highest performance was observed with a batch size of 8. The best-performing VGG16 has shown a classification accuracy of 94.33%. The proposed model has shown excellent results in the unseen data with a good level of confidence in distinguishingleaf miner infected and healthy leaves (Fig.8).

The objective behind developing an automated approach for the automatic identification of leaf miners is due to the increased destruction power of these pests in hydroponic environments. Early identification of leaf miners can allow for more targeted and cost-effective pest management with minimal use of chemicals, thus, reducing the overall environmental impact. However, this study is limited to binary classification i.e., classification into healthy and leaf miner classes. This study can be further extended by adding other classes such as such as mite, jassid, etc.

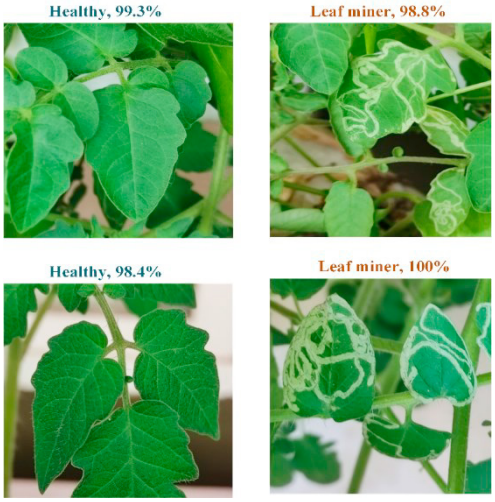


Fig 8. Classification results of attention based CNN model on the test dataset

5. Conclusion

This study introduced an approach for the identification of leaf miner pest infestation in tomatoes, which is a growing concern for tomato growers in protected cultivation systems like hydroponics. For accurate identification of the pest infestation, the present study developed an attention-based CNN model that consists of a CBAM module integrated into the custom CNN architecture. In this regard, an image dataset containing images of healthy and leaf miner infestations was created and further, the CNN model was evaluated on this dataset. The accuracy of the proposed model was obtained as 97.87%. The different pre-trained CNN models (AlexNet, VGG16, and VGG19) were also evaluated and compared with the performance of the developed model. The experimental results indicated that the proposed model is superior to these pre-trained models. Thus, the CBAM-integrated CNN model can give more attention to the infection patterns, improving the feature extraction and classification efficiency. The potential future work includes expanding the study by incorporating other pests such as mites, jassids, etc. The developed model is effective for deployment on mobile devices and other resource-constrained embedded devices.

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